

1) a) 28 b) $\frac{1}{8}$ c) 1 d) $\frac{1}{2}$ e) 216
f) 20 g) $\frac{1}{4}$ h) 11 i) $\frac{7}{8}$ j) $\frac{1}{16}$

2)

$$\begin{aligned} a_n &:= 7n, n \in \mathbb{N}^+ \\ b_n &:= \frac{1}{2n}, n \in \mathbb{N}^+ \\ c_n &:= -1^{n+1}, n \in \mathbb{N}^+ \\ d_n &:= -1^n \cdot \frac{1}{2n}, n \in \mathbb{N}^+ \\ e_n &:= n^3, n \in \mathbb{N}_0 \end{aligned}$$

$$\begin{aligned} f_n &:= \begin{cases} n=0: -10 \\ n \geq 1: f_{n-1} - n \end{cases}, n \in \mathbb{N}_0 \\ g_n &:= \text{Primzahlen} \\ h_n &:= \frac{2n-1}{2n}, n \in \mathbb{N}^+ \\ i_n &:= 2^{-n}, n \in \mathbb{N}_0 \\ j_n &:= 2^{-n}, n \in \mathbb{N}_0 \end{aligned}$$

$$f_4 = f_3 - 4 = (f_2 - 3) - 4 = ((f_1 - 2) - 3) - 4 = ((f_0 - 1) - 2) - 3 - 4 = -10 - 1 - 2 - 3 - 4$$

3)

- a) Streng monoton steigend f) Streng fallend
b) monoton fallend g) /
c) nicht monoton h) Streng monoton steigend
d) nicht monoton i) monoton steigend / streng steigend
e) Streng monoton steigend j) monoton fallend

4)

a) div f) div
b) kon $\rightarrow 0$ g) nie
c) nie h) div
d) nie i) kon $\rightarrow 1$
e) div j) kon $\rightarrow 0$

Wachstumsfolge: (nicht streng) $\begin{cases} n < 3: n \\ n \geq 4: 4 \end{cases}$

Ü2

1a)

$$\sum_{i=0}^5 ((-i) + 4)$$

$$= -0 + 4 + (-1) + 4 + (-2) + 4 + (-3) + 4 + (-4) + 4 + (-5) + 4$$

$$= 4 + 3 + 2 + 1 + 0 - 1$$

$$= 9$$

1b)

$$\prod_{i=1}^4 2i = 2 \cdot 1 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 2 \cdot 4 = 2 \cdot 4 \cdot 6 \cdot 8$$

$$= 384$$

1c)

$$n \text{ Cr } r = \frac{n!}{r! \cdot (n-r)!}$$

$$= 161700$$

2b)

$$A(n) = \sum_{i=1}^n \frac{1}{i^2 \cdot i} = 1 - \frac{1}{n+1}$$

Induktionsanfang: $n=1$

$$\sum_{i=1}^1 \frac{1}{i^2 \cdot i} = 1 - \frac{1}{n+1}$$

$$\frac{1}{1^2 \cdot 1} = 1 - \frac{1}{1+1}$$

$$\frac{1}{2} = 1 - \frac{1}{2} = \frac{1}{2} \quad /$$

Induktionsvoraussetzung: $\exists n \in \mathbb{N} : \sum_{i=1}^n \frac{1}{i^2 \cdot i} = 1 - \frac{1}{n+1}$

Induktionsschritt: $n \mapsto n+1$

$$\sum_{i=1}^{n+1} \frac{1}{i^2 \cdot i} = \sum_{i=1}^n \frac{1}{i^2 \cdot i} + \frac{1}{(n+1)^2 \cdot (n+1)}$$

$$1 - \frac{1}{(n+1) \cdot 1} \stackrel{IV}{=} 1 - \frac{1}{n+1} + \frac{1}{(n+1)^2 \cdot (n+1)}$$

$$1 - \frac{1}{n+1} = 1 - \frac{1}{n+1} + \frac{1}{n^2 \cdot 2n+1 \cdot n+1}$$

$$1 - \frac{1}{n+1} = 1 - \frac{1}{n+1} + \frac{1}{n^2+3n+2} \quad | -1 | + \frac{1}{n+1}$$

$$\frac{1}{n+1} - \frac{1}{n+1} = \frac{1}{n^2+3n+2}$$

$$\frac{n+2}{n+1} - \frac{1}{n+1} = \dots$$

$$\frac{n+2}{n+1} - \frac{1}{n+1} = \dots \quad | \cdot (n+1)$$

$$\frac{n+2 - n - 1}{n+1} = \frac{n+1}{n^2+3n+2} \quad | \cdot (n+2)$$

$$n+2 - n - 1 = \frac{n^2+3n+2}{n^2+3n+2}$$

$$1 = 1 \quad \checkmark \quad \text{Auss. 4)}$$

$$f_4 = f_3 - 4$$

2a)

Induktionsanfang: $n=1$

$$2 \cdot 1 - 1 = 1^2$$

$$1 = 1 \quad \checkmark$$

Induktionsvoraussetzung: $n \in \mathbb{N} : \sum_{i=1}^n (2i-1) = n^2$

Induktionsschritt: $n \mapsto n+1$

$$\sum_{i=1}^{n+1} (2i-1) = (n+1)^2$$

$$\sum_{i=1}^{n+1} (2i-1) = \sum_{i=1}^n (2i-1) + (2(n+1)-1)$$

$$(n+1)^2 \stackrel{IV}{=} n^2 + 2n + 1$$

$$n^2 + 2n + 1 \stackrel{IV}{=} n^2 + 2n + 1 \quad \checkmark$$

2c)

$$A(n) = \sum_{i=1}^n 4 \cdot i^3 = n^4 + 2n^3 + n^2$$

Induktionsanfang: $n=1$

$$\sum_{i=1}^1 4 \cdot i^3 = n^4 + 2n^3 + n^2$$

$$4 \cdot 1^3 = 1^4 + 2 \cdot 1^3 + 1^2$$

$$4 = 4 \quad \checkmark$$

Induktionsvoraussetzung: $\exists n \in \mathbb{N} : \sum_{i=1}^n 4 \cdot i^3 = n^4 + 2n^3 + n^2$

Induktionsschritt: $n \mapsto n+1$

$$\sum_{i=1}^{n+1} 4 \cdot i^3 = \sum_{i=1}^n 4 \cdot i^3 + 4 \cdot (n+1)^3$$

$$= \dots + 4 \cdot (n+1) \cdot (n^2 + 2n + 1)$$

$$= \dots + 4 \cdot (n^3 + 2n^2 + n + n^2 + 2n + 1)$$

$$(n+1)^4 + 2(n+1)^3 + (n+1)^2 \stackrel{IV}{=} n^4 + 4n^3 + 6n^2 + 4n + 1 + 2n^3 + 6n^2 + 6n + 2 + n^2 + 2n + 1$$

$$(n+1)^4 + 2(n+1)^3 + (n+1)^2 \stackrel{IV}{=} n^4 + 2n^3 + n^2 + 4n^3 + 12n^2 + 12n + 4$$

$$n^4 + 3n^3 + 3n^2 + n + n^3 + 3n^2 + 3n + 1 + 2n^3 + 6n^2 + 6n + 2 + n^2 + 2n + 1 = \dots$$

$$n^4 + 6n^3 + 13n^2 + 12n + 4 = n^4 + 6n^3 + 13n^2 + 12n + 4 \quad \checkmark$$

Auss. 3)

$$A(n) : 2^n > n, n \in \mathbb{N}_0$$

2^n skaliert schneller als n
und hat ein höheres Startgsw.
für $n \in \mathbb{N}_0$

IA: $2^0 > 0 \Rightarrow 1 > 0 \quad \checkmark$

IV: $\exists n \in \mathbb{N}_0 : 2^n > n$

IS: $2^{n+1} > n+1$

$$2 \cdot 2^n > 2n \geq n+1$$

4)

$$A(n) : \exists n \in \mathbb{N} : 7^n - 1 = 6 \cdot 2_n$$

IA: $n=1$

$$7^1 - 1 = 6 \cdot 2_n \quad | :6$$

$$1 = 2_n, 1 \in \mathbb{N}^+ \quad \checkmark$$

IV: $A(n)$

IS: $n \mapsto n+1 \quad \frac{a}{b} \mid \frac{c}{d} \rightarrow \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \neq 0$

$$7^{n+1} - 1 \mid 6$$

$$7^n \cdot 7 - 1 \mid 6$$

$$6 \cdot 7^n, 7^n - 1 \mid 6$$

$$6 \cdot 7^n \mid 6$$

$$7^n \mid 1$$

da $7^n \in \mathbb{N} : 7^n = 2_n$

d.h.: $6 \cdot 2_n \% 7^n - 1 = 0 \quad \checkmark$

| IV = Produktregel, wenn IV $\% 6 \neq 0$, dann

$$| :6$$

3)

1) a) $a_n := \frac{25}{n^2}$
 $a := \lim_{n \rightarrow \infty} a_n = 0$
 $\varepsilon = 0,1$
 $(n \geq n_\varepsilon) \rightarrow (|a_n - a| < \varepsilon)$
 $\frac{25}{n^2} < 0,1 \quad | \cdot n^2 : 10$
 $250 < n^2 \quad | \sqrt{\quad}$
 $\sqrt{250} < n \rightarrow n_\varepsilon = 16$
 $(15,81 \geq 15) \rightarrow (|\frac{25}{n^2} - 0| < 0,1)$
für $\varepsilon = 0,01$
 $\frac{25}{n^2} < 0,01 \quad | \cdot n^2 : 100 \sqrt{\quad}$
 $50 < n$
 $(50 \geq 50) \rightarrow (|\frac{25}{n^2} - 0| < 0,01)$
 $n \rightarrow 51$
 $(51 \geq 51) \rightarrow (|\frac{25}{n^2} - 0| < 0,01)$

b) $b_n = \frac{2n-7}{\sqrt{2}n+1}$
 $b = \lim_{n \rightarrow \infty} b_n = \sqrt{2}$
 $(n \geq n_\varepsilon) \rightarrow (|b_n - b| < \varepsilon)$
für $\varepsilon = 0,1$:
 $\frac{2n-7}{\sqrt{2}n+1} < \sqrt{2} - 0,1$

4)
a) div b) div c) kon

1)
a) 0 b) 1 c) e
d) $\infty \cdot \frac{1}{\infty} = 1$

3)
c) 1 b) 0 c) ∞
d) 0 e) $\frac{10}{3}$ f) ∞^{10}
g) $\infty^{10}, -\infty^{10}, 0$
h) ∞^{10}

weitere Info = 1 Punkt (Klausur)

Apf. machen (Gut) }
return Apf. geben }

4)
1) a) $7 \in \mathbb{N}_0, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$
b) $\sqrt{7} \in \mathbb{R}$
c) $\sqrt{-3} \in \mathbb{I}$
d) $\sqrt{\frac{4}{5}} \in \mathbb{Q}, \mathbb{R}$
e) $-55 \in \mathbb{Z}, \mathbb{G}, \mathbb{R}$
f) $\pi \in \mathbb{R}$

2)
a) $\frac{1}{2}$ b) $T \in X$
 $[-11; -1,5], [0,5; 2] \in T$
 $[-11; -1,5] \cup [0,5; 2] \notin Y$

3)
a) True
b) True, das Supremum ist $+\infty$
c) True
d) Falsch

4)
 $f(x) = 0$
 $0 = \frac{x^2-1}{x^2-4} \quad | \text{Satz des NP}$
Wenn $p = 0$ bei $\frac{p}{q}$
dann: $\frac{p}{q} = 0$

$0 = x^2 - 1 \quad | -1$
 $1 = x^2 \quad | \sqrt{\quad}$
 $x_1 = 1$
 $x_2 = -1$

Definitionsbereich:
 $x^2 - 4 \neq 0 \quad | +4 \quad | \sqrt{\quad}$
 $\mathbb{D} = \mathbb{R} \setminus \{2, -2\}$
Polstellen:

$\lim_{x \rightarrow 2^+} \frac{x^2-1}{x^2-4} = \infty$
 $\lim_{x \rightarrow 2^-} \frac{x^2-1}{x^2-4} = -\infty$

daraus folgt: Polstelle bei $x = 2, -2$

b) $f(x) = \frac{x^2-2x-8}{x^2-5}$
 $f(x) = 0$
 $0 = \frac{x^2-2x-8}{x^2-5} \quad | \text{Satz des NP}$

$x^2-2x-8 = 0 \quad | p_q$
 $-1 \pm \sqrt{1+8} = 0$
 $x_1 = 2$
 $x_2 = 4$
Definitionsbereich:
 $x^2-5 = 0 \quad | +5 \quad | \sqrt{\quad}$
 $x_1 = 3$
 $x_2 = -3$

Polstellen:
 $\lim_{x \rightarrow 3^+} f(x) = \infty$
 $\lim_{x \rightarrow 3^-} f(x) = -\infty$
 $\lim_{x \rightarrow -3^+} f(x) = \infty$
 $\lim_{x \rightarrow -3^-} f(x) = -\infty$
Polstellen bei: $x_1 = 3$
 $x_2 = -3$

c) $f(x) = \frac{x+1}{x^2+4x+4}$
Natr. Bed.
 $f(x) = 0$
 $0 = \frac{x+1}{x^2+4x+4} \quad | \text{Satz des NP}$
 $0 = x+1 \quad | -1$

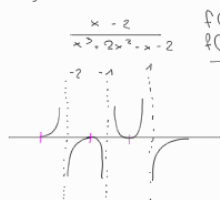
$\underline{x = -1}$
Definitionsbereich:
 $x^2+4x+4 = 0 \quad | p_q$
 $-2 \pm \sqrt{4-4} = x$
 $\underline{x = -2} \quad \mathbb{D} = \mathbb{R} \setminus \{-2\}$

Polstellen:
 $\lim_{x \rightarrow -2^+} \frac{x+1}{x^2+4x+4} = -\infty$
 $\lim_{x \rightarrow -2^-} \frac{x+1}{x^2+4x+4} = \infty$
Polstelle bei: $x = -2$

d) $f(x) = \frac{x-2}{x^2+2x^2-x-2}$
 $f(x) = 0$
 $0 = \frac{x-2}{x^2+2x^2-x-2} \quad | \text{Satz des NP}$
 $0 = x-2 \quad | +2$
 $\underline{x = 2}$

Definitionsbereich:
 $x^2-2x^2-x-2 = 0$
 $(x^2-1)(x-2) = 0 \quad | \text{Satz des NP}$
 $x^2-1 = 0 \quad | +1 \quad | \sqrt{\quad}$
 $x_1 = 1$
 $x_2 = -1$
 $x-2 = 0 \quad | +2$
 $x_3 = 2$
 $\mathbb{D} = \mathbb{R} \setminus \{1, -1, 2\}$

Polstellen:
 $\lim_{x \rightarrow 1^+} f(x) = -\infty$
 $\lim_{x \rightarrow 1^-} f(x) = \infty$
 $\lim_{x \rightarrow 2^+} f(x) = -\infty$
 $\lim_{x \rightarrow 2^-} f(x) = \infty$
 $\lim_{x \rightarrow -1^+} f(x) = \infty$
 $\lim_{x \rightarrow -1^-} f(x) = -\infty$
Polstellen bei: $-2, 1, -1$



$f(-x) \neq -f(x)$
 $f(-x) \neq f(x)$
 $\rightarrow$ keine Symmetrie

e) $f(x) = \frac{x^3 - x^2 + 3x - 3}{x^2 - x - 2}$

Nehr. Bed.: $f(x) = 0$
 $0 = \frac{x^3 - x^2 + 3x - 3}{x^2 - x - 2}$ | Satz des NP

$0 = x^3 - x^2 + 3x - 3$
 $0 = (x^2 + 3)(x - 1)$ | Satz des NP

$x^2 + 3 = 0$ | -3 $x - 1 = 0$ | +1
 $x^2 = -3$ | $\sqrt{-1}$ $x_1 = 1$
 $x_1 = 3i$
 $x_2 = -3i$

Definitionslücken:

wenn $q=0$ bei $\frac{p}{q}$
 dann Def. Lücke

$\rightarrow 0 = x^2 - x - 2$ | p4
 $x_1 = 2$
 $x_2 = -1$

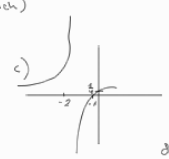
$D = \mathbb{R} \setminus \{-1, 2\}$

Polstellen:
 $\lim_{x \rightarrow -1^-} f(x) = -\infty$
 $\lim_{x \rightarrow -1^+} f(x) = \infty$
 $\lim_{x \rightarrow 2^-} f(x) = \infty$
 $\lim_{x \rightarrow 2^+} f(x) = -\infty$
 $\rightarrow$ Polstellen bei $\{-1, 2\}$

V

Verhalten schon vorhanden (glaube ich)

Symmetrie:
 $\rightarrow$ y-Achsen-Symmetrie auf $x=0$



b) 3
 $\hookrightarrow$ Punkt Symmetrie auf $(0, \frac{8}{3})$

e) $f(x) = \frac{x^3 - x^2 + 3x - 3}{x^2 - x - 2}$

$f(-x) = \frac{-x^3 - x^2 - 3x - 3}{x^2 - x - 2} \neq f(x) \vee -f(x) \Rightarrow$ keine Symmetrie

f) $f(x) = \frac{x^6 - x^4 + 2x^2 - 2}{x^4 + 4x^2 + 3}$

$f(-x) = \frac{x^6 - x^4 + 2x^2 - 2}{x^4 + 4x^2 + 3} = f(x) \Rightarrow$ Achsensymmetrie

VI

a) $2 \sin(\frac{1}{5}x + 0.3) = 1.6$
 $2 \sin(\frac{1}{5}x + 0.3 + 2\pi \cdot n) = 1.6, n \in \mathbb{Z}$ | $\div 2$ | $\arcsin()$

$\frac{1}{5}x + 0.3 + 2\pi \cdot n = \arcsin(0.8) \quad | \cdot 0.3 | \cdot 2\pi \cdot n \quad | \cdot 5$
 $x = 5 \arcsin(0.8) - 1.5 + 10\pi \cdot n, n \in \mathbb{N}$
 $\forall n, n \in \mathbb{Z} \left\{ \begin{array}{l} x_1 = 5 \arcsin(0.8)_1 - 1.5 + 10\pi \cdot n \\ x_2 = 5 \arcsin(0.8)_2 - 1.5 + 10\pi \cdot n \end{array} \right\}$

b) $\cos(-7x) = -\frac{1}{4}$
 $\cos(-7x + 2\pi \cdot n) = -\frac{1}{4}$ | $\arccos()$
 $-7x + 2\pi \cdot n = \arccos(-\frac{1}{4}) \quad | \cdot -1 \quad | \cdot 2\pi \cdot n \quad | \div 7$
 $\forall n, n \in \mathbb{Z} \left\{ \begin{array}{l} x_1 = \frac{\arccos(-\frac{1}{4}) + 2\pi \cdot n}{-7} \\ x_2 = \frac{\arccos(-\frac{1}{4}) - 2\pi \cdot n}{-7} \end{array} \right\}$

c) $\frac{1}{3} \tan(\frac{1}{5}x + 0.3) = 1$ | $\cdot 3$
 $\tan(\frac{1}{5}x + 0.3) = 3$
 $\tan(\frac{1}{5}x + 0.3 + 2\pi \cdot k) = 3$ | $\arctan()$
 $\frac{1}{5}x + 0.3 + 2\pi \cdot k = \arctan(3) \quad | \cdot 5 \quad | \cdot -0.3 \quad | \cdot 2\pi \cdot k \quad | \cdot 5$
 $x = 5 \arctan(3) - 1.5 + 10\pi \cdot k, k \in \mathbb{Z}$

e) $\sin(6x - 2) - \cos(6x - 2) = 0$ | $\cdot \cos(6x - 2)$
 $\sin(6x - 2) = \cos(6x - 2)$
 aus $\sin(\theta) = \cos(\theta)$, folgt: $\theta = \frac{1}{4}\pi + k\pi, k \in \mathbb{Z}$
 $6x - 2 = \frac{1}{4}\pi + k\pi$ | $\cdot 2$ | $\cdot \frac{1}{6}$
 $x = \frac{\frac{1}{4}\pi + k\pi + 2}{6}, k \in \mathbb{Z}$

g) $f(x) = \frac{1}{\ln(x)}$

Nehr. Bed.: $f(x) = 0$
 $0 = \frac{1}{\ln(x)}$ | Satz des NP
 $0 = 1$ $\nexists$
 $\rightarrow f(x)$ hat keine Nullstellen

Definitionslücken:
 $\ln(x) = 0$ | $x = 1$
 $e^{\ln(x)} = e^0$
 $x = 1$
 $D = \mathbb{R} \setminus \{1\}$

Polstellen:
 $\lim_{x \rightarrow 1^+} \frac{1}{\ln(x)} = \infty$
 $\lim_{x \rightarrow 1^-} \frac{1}{\ln(x)} = -\infty$
 $\rightarrow$ Polstelle bei $x=1$

h) $f(x) = (2x^5 - 64x) \cdot \sin(x)$

Nehr. Bed.: $f(x) = 0$
 $0 = (2x^5 - 64x) \cdot \sin(x)$ | Satz des NP

$x \cdot (2x^4 - 64) = 0$ | Satz des NP $\sin(x) = 0$ | $\arcsin()$
 $2x^4 - 64 = 0$ | $\div 2$ | $x_2 = 0$
 $x^4 = 32$ | $\sqrt[4]{}$
 $x_1 \approx 2.38$
 $x_2 = 0$
 $x_3 \approx -2.38$
 $x_4 \approx 2.38i$
 $x_5 \approx -2.38i$

Definitionslücken:
 $\sin(x)$ hat keine Definitionslücken
 $2x^5 - 64x$ hat auch keine "
 daraus folgt: $D = \mathbb{R}$, $f(x)$ hat also keine DL.

Polstellen:
 Auch keine, da $D = \mathbb{R}$

j) $f(x) = \frac{1}{\ln(x)}$
 $f(-x) = \frac{1}{\ln(-x)} = \frac{1}{\ln(x) \cdot \frac{\pi i}{2}} \neq f(x) \vee -f(x)$
 $\rightarrow$ keine Symmetrie

h) $f(x) = (2x^5 - 64x) \cdot \sin(x)$
 $f(-x) = (-2x^5 + 64x) \cdot \cos(x + \frac{\pi}{2})$
 $f(-x) \neq f(x) \vee -f(x) \Rightarrow$ keine Symmetrie

- 2)
- $f_1(x)$ = gebrochen-rational
 - $f_2(x)$ = ganzrational
 - $f_3(x)$ = gebrochen-rational
 - $f_4(x)$ = trigonometrisch, $\tan(x)$
 - $f_5(x)$ = exponentiell
 - $f_6(x)$ = rational
 - $f_7(x)$ = trigonometrisch, $\sin(x)$
 - $f_8(x)$ = exponentiell

$2 \cdot 2^{\frac{x}{2}}$
 d) $\cot(x - \frac{1}{2}\pi) = -3$
 $\cot(x - \frac{1}{2}\pi + 2\pi \cdot k) = -3$ | $\operatorname{arccot}()$
 $x - \frac{1}{2}\pi + 2\pi \cdot k = \operatorname{arccot}(-3) \quad | \cdot \frac{1}{2}\pi \quad | \cdot 2\pi \cdot k$
 $x = \operatorname{arccot}(-3) + \frac{1}{2}\pi + 2\pi \cdot k, k \in \mathbb{Z}$

2)

$$f(x) = \sqrt{x}$$

$$f'(x) = \lim_{\delta \rightarrow 0} \frac{f(x+\delta) - f(x)}{\delta}$$

$$f'(x) = \lim_{\delta \rightarrow 0} \frac{\sqrt{x+\delta} - \sqrt{x}}{\delta} \cdot (\sqrt{x+\delta} + \sqrt{x}) \cdot (\sqrt{x+\delta} - \sqrt{x})$$

$$f'(x) = \lim_{\delta \rightarrow 0} \frac{x + \delta - x}{\delta \cdot (\sqrt{x+\delta} + \sqrt{x})}$$

$$f'(x) = \lim_{\delta \rightarrow 0} \frac{1}{\sqrt{x+\delta} + \sqrt{x}}$$

$$f'(4) = \frac{1}{\sqrt{4+0} + \sqrt{4}}$$

$$f'(4) = \frac{1}{4}$$

$$\sqrt{x}' = \frac{1}{2\sqrt{x}} \cdot x'$$

3) a) $f'(u) = 1 - \cos(x)$

b) $f'(u) = \sqrt{3} \cdot e^x - 2 \sin\left(\frac{\pi}{6}x\right) \cdot \frac{\pi}{6}$

c) $f'(u) = \sqrt{e^{40}} = e^{20} \Rightarrow f'(u) = 40e^{20}$

d) $f(x) = 4 \ln(x) + 2x^3 + 5$

$$f'(x) = \frac{4}{x} + 6x^2$$

e) $f(x) = u \cdot v + c = x^2 \cdot \sin(x) - e^{\frac{1}{2}}$

$$f'(x) = u' \cdot v + u \cdot v' + c' = 2x \cdot \sin(x) + x^2 \cdot \cos(x) - \frac{e^{\frac{1}{2}}}{2}$$

f) $f(x) = x \cdot \frac{\sqrt{4x-1}}{x-1}$

$$= x \cdot (4x-1)^{\frac{1}{2}} \cdot (x-1)^{-1}$$

$$= u \cdot v = u \cdot (u' \cdot n)$$

$$u = x$$

$$u' = \sqrt{4x-1}$$

$$n = (x-1)^{-1}$$

$$f'(x) = u' \cdot v + u \cdot v'$$

$$= u' \cdot v + u \cdot (u' \cdot n + u \cdot n')$$

$$u' = 1$$

$$u' = \frac{1}{2 \cdot \sqrt{4x-1}} \cdot 4$$

$$n' = (x-1)^{-2} = \frac{1}{(x-1)^2} = \frac{1}{x^2-2x+1}$$

$$= 1 \cdot \frac{\sqrt{4x-1}}{x-1} + x \cdot \left(\frac{4}{2 \cdot \sqrt{4x-1}} \cdot \frac{1}{x-1} + \frac{\sqrt{4x-1}}{x^2-2x+1} \right)$$

$$= \frac{\sqrt{4x-1}}{x-1} + \frac{4x}{(2 \cdot \sqrt{4x-1}) \cdot (x-1)} + \frac{\sqrt{4x-1}}{x^2-2x+1}$$

$$= \frac{\sqrt{4x-1}}{x-1} + \frac{4x}{2x-2 + x\sqrt{4x-1} - \sqrt{4x-1}} + \frac{\sqrt{4x-1}}{x^2-2x+1}$$

$$= \frac{x\sqrt{4x-1} - \sqrt{4x-1}}{x^2-2x+1} + \frac{4x}{x^2-2x+1} + \frac{\sqrt{4x-1}}{x^2-2x+1}$$

$$= \frac{x \cdot \sqrt{4x-1}}{x^2-2x+1} + \frac{4x}{2x-2 + x\sqrt{4x-1} - \sqrt{4x-1}} \quad ; \text{ numerator } ; \text{ Zähler}$$

$$= \frac{1}{x^2-2x+1}$$

3) $f(u) = (u+x')^2 = (x^2+u) \cdot (x^2+u)$

$$f(x) = x^4 + 2ux^2 + u^2$$

$$f'(u) = 4x^2 + 4ux$$

$$f'(u) = 2u + 2x^2$$

$$\text{Qu. e.: } f(x) = \frac{u}{v}, f'(x) = \frac{u' \cdot v - u \cdot v'}{v^2}$$

$$f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$f'(x) = \frac{\cos(x) \cdot \cos(x) - \sin(x) \cdot (-\sin(x))}{(\cos(x))^2} = \frac{\cos(x)^2 + \sin(x)^2}{\cos(x)^2}$$

$$\rightarrow \text{Trig. Pythagoras: } \sin(x)^2 + \cos(x)^2 = 1$$

$$\hookrightarrow f'(x) = \frac{1}{\cos(x)^2}$$

$$f''(x) = \frac{0 \cdot \cos(x)^2 + 1 \cdot [\cos(x)^2]'}{\cos(x)^4}$$

$$[\cos(x)^2]' = \cos(x) \cdot (-\sin(x)) + (-\sin(x)) \cdot \cos(x)$$

$$= -2 \cos(x) \sin(x)$$

$$f''(x) = \frac{-2 \cos(x) \sin(x)}{\cos(x)^4} = \frac{-2 \sin(x)}{\cos(x)^3}$$

VII Aufg. 1)

$$V = 0.001 \text{ m}^3$$

$$125 \text{ mL} = 0.125 \text{ L} = 0.000125 \text{ m}^3$$

$$V_B = \pi r^2 \cdot h$$

$$O_B = 2\pi r \cdot h + 2 \cdot \pi r^2 : \text{Mantel + Kreise}$$

Bedingung: $V_B = 0.000125 \text{ m}^3$

$$0.000125 \text{ m}^3 = \pi r^2 \cdot h \quad | : \pi r^2$$

$$\frac{125 \text{ m}^3}{1000000 \pi r^2} = h$$

h in O_B einsetzen:

using namespace m3:

$$O(r) = 2\pi r \cdot \frac{125}{1000000 \pi r^2} \cdot 2\pi r^2$$

$$O(r) = \frac{2\pi r \cdot 125}{1000000 \pi r^2} \cdot 2\pi r^2$$

$$O(r) = \frac{25}{100000} \cdot r + 2\pi r^2$$

$$O'(r) = \frac{25}{100000} + 4\pi r$$

Nchw. Bedingung: $O'(r) = 0$

$$0 = \frac{25}{100000 \cdot r^2} + 4\pi r \quad | \cdot 4\pi r$$

$$4\pi r = \frac{25}{100000 \cdot r^2} \quad | \cdot 100000 \cdot r^2$$

$$40000\pi r^3 = 25 \quad | : 40000$$

$$r^3 = \frac{25}{40000\pi} \quad | \sqrt[3]{}$$

$$r = 0.0271 \text{ m} = 2.71 \text{ cm}$$

$$h = \frac{125 \text{ m}^3}{1000000 \pi \cdot (0.0271 \text{ m})^2}$$

$$h = 5.42 \text{ cm}$$

②

$$x \in \mathbb{R}$$

a) $\mathcal{L} = \{x \in \mathbb{R} \mid \frac{1}{2x+5} \geq -4 \mid \wedge (-1)$

b) $\mathcal{L} = \{x \in \mathbb{R} \mid -\frac{2}{3} \leq x \leq \frac{3}{5}\}$

c) $\left| \frac{5x}{15} - \frac{3}{15} \right| \leq -\frac{1}{15} \quad x \leq -\frac{1}{8} - \frac{3}{2}$

$$\mathcal{L} = \{ \}, \text{ da der Absolutbetrag von egal welcher Reellen Zahl nicht kleiner als 0 werden kann.}$$

$$\mathcal{L} = \{x \in \mathbb{R} \mid x \leq -\frac{21}{8}\}$$

$$P: u \cdot v = \int v \cdot u' dx$$

VIII

1)

a)

$$x^3 + 1/x + C, C \in \mathbb{R}$$

b) $\int \ln(x) \cdot 1 dx$

$$\ln(x) \cdot x = \int x \cdot \frac{1}{x} = x \ln(x) - x + C, C \in \mathbb{R}$$

c) $\int f(x) dx = \frac{1}{4} x^4 - \frac{1}{3} x^3 + \frac{1}{2} x^2 + C, C \in \mathbb{R}$

d) $\int f(x) dx = \frac{1}{6} x^6 - \cos(x) + C, C \in \mathbb{R}$

e) $\int f(x) dx = -\sin(x) + C, C \in \mathbb{R}$

f) $\int f(x) dx = \int x^3 dx = \frac{x^4}{4} + C, C \in \mathbb{R}$

g) $\int f(x) dx = \int 2x^{-\frac{1}{2}} dx = \sqrt{x} + C, C \in \mathbb{R}$

h) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

i) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

j) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

k) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

l) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

m) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

n) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

o) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

p) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

q) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

r) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

s) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

t) $\int f(x) dx = \frac{\ln(x)}{n} + C, C \in \mathbb{R}$

$$\int_{N_1}^N (x^2 - 4x - 12)(x-2) dx = \int_{N_1}^N (x^3 - 4x^2 - 12x - x^2 + 4x + 24) dx$$

$$= \int_{N_1}^N (x^3 - 5x^2 - 8x + 24) dx = \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 - 4x^2 + 24x \right]_{N_1}^{N_2}$$

Nöw. Bedingung: $f(x) = 0$

$$0 = (x^2 - 4x - 12)(x-2) \quad | : (x-2)$$

$$x^2 - 4x - 12 = 0 \quad | +4$$

$$x_1 = 2 + \sqrt{16} = 6$$

$$x_2 = 2 - \sqrt{16} = -2$$

$$A = \left| \int_2^6 f(x) dx \right| + \left| \int_{-2}^2 f(x) dx \right| = |164| + |164| = 328 \text{ FE}$$

$$f(x) = \frac{4x^3}{x^3 + 2x^2 - x - 2}, F(x) = 4 \cdot \int \frac{x^3}{x^3 + 2x^2 - x - 2} dx$$

NS des Nenners:

$$x^3 + 2x^2 - x - 2 = 0$$

$$(x^2 - 1)(x + 2) = 0 \quad | \text{ Satz des Nullprodukts}$$

$$x^2 - 1 = 0 \quad | +1 \quad | \sqrt{\quad}$$

$$x = \pm 1$$

$$\Rightarrow x_1 = 1, x_2 = -1$$

$$\Rightarrow \text{NS } (1, -1, -2)$$

$$\Rightarrow f(x) = \frac{4x^3}{(x-1)(x+1)(x+2)} \quad | \text{ Partialbruchzerlegung}$$

$$\frac{-2x^2 + x + 2}{(x-1)(x+1)(x+2)} = \frac{a}{x-1} + \frac{b}{x+1} + \frac{c}{x+2}$$

$$= a(x+1)(x+2) + b(x-1)(x+2) + c(x-1)(x+1)$$

$$= a(x^2 + 3x + 2) + b(x^2 + x - 2) + c(x^2 - 1)$$

$$= ax^2 + 3ax + 2a + bx^2 + bx - 2b + cx^2 - c$$

$$= x^2(a+b+c) + x(3a+b) + (2a-2b-c) \quad | \text{ Koeffizientenvergleich}$$

$$\Rightarrow \text{I } a+b+c = -2$$

$$\text{II } 3a+b = 1 \quad | -3a$$

$$\text{III } 2a-2b-c = 2$$

$$\text{II}' \text{ in I:}$$

$$\text{I}' a + 1 - 3a + c = -2 \quad | +1$$

$$-2a + c = -3 \quad | +2a$$

$$\text{II}' \text{ in III:}$$

$$\text{III}' 2a - 2(1-3a) - (-3+2a) = 2$$

$$2a - 2 + 6a + 3 - 2a = 2 \quad | :1$$

$$6a = 1 \quad | :6$$

$$a = \frac{1}{6}$$

$$a \text{ in II':}$$

$$\text{II}'' b = 1 - \frac{1}{6} = \frac{5}{6}$$

$$a \text{ in I':}$$

$$\text{I}'' c = -3 + \frac{1}{6} = -\frac{17}{6}$$

$$\Rightarrow a = \frac{1}{6}, b = \frac{5}{6}, c = -\frac{17}{6}$$

$$a, b, c \text{ in } F(x) = 4 \int \frac{a}{x-1} + \frac{b}{x+1} + \frac{c}{x+2} dx$$

$$F(x) = 4 \left(\int \frac{1}{2x-2} dx + \int \frac{1}{6x+6} dx - \int \frac{17}{6x+12} dx + \int 1 dx \right)$$

$$F(x) = 4 \left(\ln(12x-12) + \ln(6x+6) - \ln\left(\frac{17}{6}x+2\right) + x \right)$$

4)

$$f(x) = x^2 - 4x$$

$$g(x) = -\frac{1}{2}x^2 + 2x$$

Nöw. Bed.: $f(x) = 0$ & $g(x) = 0$:

$$f(x) = 0 = x^2 - 4x$$

$$x_1 = 0, x_2 = (0 + 4) = 4$$

$$g(x) = 0 = -\frac{1}{2}x^2 + 2x$$

$$x_1 = 0$$

$$x_2 = \left(-\frac{1}{2}x + 2 = 0\right) = 4$$

$$\int_{-3}^1 f(x) dx - \int_{-3}^1 g(x) dx \approx 35,2$$

$$c) f(x) = e^{-\sin(x)} \cdot \cos(x)$$

$$u = -\sin(x) \Rightarrow \frac{du}{dx} = -\cos(x) \quad | \cdot dx \quad | \cdot \frac{1}{-\cos(x)}$$

$$dx = \frac{du}{-\cos(x)}$$

$$= \int e^u du = e^u = e^{-\sin(x)}, C$$

$$e) f(x) = \frac{1}{x^2 - n^2} \cdot n^2 + 0$$

$$f(x) = \frac{1}{x^2 - n^2}$$

NS von $x^2 - n^2$:

$$x^2 - n^2 = 0 \quad | + n^2 \quad | \sqrt{\quad}$$

$$x = \pm n \Rightarrow x_1 = n, x_2 = -n$$

$$\Rightarrow f(x) = \frac{1}{(x-n)(x+n)} \quad | \text{ Partialbruchzerlegung}$$

$$\Rightarrow \frac{a}{x-n} + \frac{b}{x+n} = \frac{1}{(x-n)(x+n)} \quad | \cdot (x-n)(x+n)$$

$$a(x+n) + b(x-n) = 0x + 1$$

$$ax + an + bx - bn = 0x + 1$$

$$(a+b)x + an - bn = 0x + 1 \quad | \text{ Koeffizientenvergleich}$$

$$\Rightarrow \text{I } a+b = 0 \quad | -b$$

$$\Rightarrow \text{II } an - bn = 1$$

$$\text{I}' \text{ in II eingesetzt:}$$

$$\text{II}' -bn - bn = -2bn = 1 \quad | \cdot \left(-\frac{1}{2n}\right)$$

$$b = -\frac{1}{2n}$$

$$\text{II}'' \text{ in I' eingesetzt:}$$

$$\text{I}'' a = \frac{1}{2n}$$

$$\Rightarrow \frac{1}{x^2 - n^2} = \frac{1}{2n(x-n)} - \frac{1}{2n(x+n)}$$

$$\Rightarrow F(x) = \frac{1}{2n} \int \frac{1}{x-n} dx - \frac{1}{2n} \int \frac{1}{x+n} dx$$

$$= \frac{1}{2n} \cdot \ln(|x-n|) - \frac{1}{2n} \cdot \ln(|x+n|)$$

$$F(x) = \frac{\ln(|x-n|) - \ln(|x+n|)}{2n} + C$$

$$j) f(x) = x \cdot \sqrt{x+1}$$

$$F(x) = \int x \cdot \sqrt{x+1} dx$$

$$\text{Substitution: } u = x+1$$

$$\Rightarrow x = u-1$$

$$\Rightarrow F(x) = \int (u-1) \cdot \sqrt{u} du$$

$$F(x) = \int u^{\frac{3}{2}} - \sqrt{u} du$$

$$F(x) = \int u^{\frac{3}{2}} du - \int u^{\frac{1}{2}} du$$

$$= \frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \quad | u = x+1$$

$$F(x) = \frac{2(x+1)^{\frac{5}{2}}}{5} - \frac{2(x+1)^{\frac{3}{2}}}{3} + C$$

$$\textcircled{X} a) f(x) = \cos(2x+u) \quad | \text{ Kettenregel}$$

$$F(x) = \frac{\sin(2x+u)}{2} + C \quad \Rightarrow \text{Wertwert der inneren Abl.}$$

$$d) f(x) = x \cdot \ln(x)$$

$$F(x) = u \cdot v - \int v \cdot u' dx$$

$$u = \ln(x), u' = \frac{1}{x}$$

$$v' = x, v = \frac{1}{2}x^2$$

$$F(x) = \ln(x) \cdot \frac{1}{2}x^2 - \int \frac{1}{2}x^2 \cdot \frac{1}{x} dx$$

$$= \dots - \int \frac{1}{2}x dx$$

$$F(x) = \frac{\ln(x) \cdot x^2}{2} - \frac{1}{4}x^2 + C$$

$$b) f(x) = (5x-10)^{23}$$

$$F(x) = \int (5x-10)^{23} dx$$

$$F(x) = \frac{1}{u'} \int u^{23} du + C$$

$$u = 5x-10$$

$$\Rightarrow F(u) = \frac{1}{u'} \int u^{23} du = \frac{1}{24} u^{24} \quad | u \Rightarrow x \quad | \cdot \frac{1}{\text{innerer Wert}}$$

$$\Rightarrow F(x) = \frac{\frac{1}{24}(5x-10)^{24}}{5} + C$$

$$h) f(x) = (\cos(2x+4))^2$$

$$F(x) = \int (\cos(2x+4))^2 dx \quad | a = 2x+4$$

$$\cos^2(a) = \frac{1 + \cos(2a)}{2}$$

$$F(x) = \int \frac{1 + \cos(2a)}{2} da$$

$$F(x) = \int \frac{1}{2} da + \int \frac{\cos(2a)}{2} da$$

$$\int \frac{1}{2} da = \frac{1}{2}a$$

$$\int \frac{1}{2} \cos(2a) da = \frac{1}{4} \sin(2a)$$

$$2a = 2(2x+4) = 4x+8$$

$$\Rightarrow F(x) = \frac{a}{2} + \frac{\sin(4x+8)}{4}$$

$$= \frac{2x+4}{2} + \frac{\sin(4x+8)}{4} + C$$

$$= \frac{4x+8 + \sin(4x+8)}{4}$$

$$i) f(x) = \frac{4x-2}{x^2-2x-63}$$

Nullstellen vom Nenner:

$$x^2 - 2x - 63 \quad | p q$$

$$x_1 = 9, x_2 = -7$$

$$\text{Partialbruchzerl.:} \Rightarrow \frac{4x-2}{(x-9)(x+7)} = \frac{a}{x-9} + \frac{b}{x+7} \quad | (x)$$

$$4x-2 = a(x+7) + b(x-9)$$

$$4x-2 = ax + 7a + bx - 9b$$

$$4x-2 = x(a+b) + (7a-9b) \quad | \text{ Koeffizientenvergleich}$$

$$\text{I } 4 = a+b \quad | -b$$

$$\text{II } -2 = 7a-9b$$

$$\text{I}' \text{ in II:}$$

$$-2 = 7(4-b) - 9b$$

$$-2 = 28 - 7b - 9b \quad | -28$$

$$-30 = -16b \quad | : -16$$

$$\frac{30}{16} = b$$

$$\text{II}' \text{ in I:}$$

$$\text{I}'' 4 = a + \frac{30}{16} \quad | -\frac{30}{16}$$

$$F(x) = \int \frac{34}{16} \cdot \frac{1}{x-9} dx + \int -\frac{30}{16} \cdot \frac{1}{x+7} dx$$

$$= \frac{34}{16} \int \frac{1}{x-9} dx - \frac{30}{16} \int \frac{1}{x+7} dx$$

$$F(x) = \frac{34}{16} \ln(|x-9|) - \frac{30}{16} \ln(|x+7|)$$

$$a) \int_0^{\frac{\pi}{2}} \sin(2x \cdot \frac{\pi}{3}) dx = \left[\frac{-\cos(2x \cdot \frac{\pi}{3})}{2} \right]_0^{\frac{\pi}{2}} \\ = \dots$$

$$b) \int_1^3 (\sqrt{5x+1}) dx \quad | \text{Sub.} \\ u = 5x+1 \Rightarrow du = 5dx \\ \Rightarrow \frac{1}{5} \int u^{\frac{1}{2}} du = \frac{1}{5} \cdot \frac{2}{3} u^{\frac{3}{2}} \\ = \frac{2u^{\frac{3}{2}}}{15} \quad | u = 5x+1 \\ = \frac{2(5x+1)^{\frac{3}{2}}}{15} \Rightarrow \left[\frac{2(5x+1)^{\frac{3}{2}}}{15} \right]_1^3$$

$$4) f(x) = \sqrt{7+2x} \\ T_{2,5} f(x) = a_2 x^2 + a_1 x + a_0 \\ \dots f'(x) = 2a_2 x + a_1 \\ \dots f''(x) = 2a_2 \\ \hline T_{2,5} f(5) = 81a_2 + 9a_1 + a_0 \\ \dots f'(5) = 18a_2 + a_1 \\ \dots f''(5) = 2a_2 \\ \hline f(x) = \sqrt{2x+7} \\ f'(x) = \frac{2}{2\sqrt{2x+7}} = (2x+7)^{-\frac{1}{2}} \\ f''(x) = -\frac{1}{(2x+7)^{\frac{3}{2}}}$$

$$f(5) = T_{2,5} f(5) \\ \dots = \dots \\ \sqrt{2 \cdot 5 + 7} = 81a_2 + 9a_1 + a_0 \\ \frac{2}{2\sqrt{2 \cdot 5 + 7}} = 18a_2 + a_1 \\ -\frac{1}{(2 \cdot 5 + 7)^{\frac{3}{2}}} = 2a_2 \quad | \cdot \frac{1}{2} \\ a_2 = -\frac{1}{2(2 \cdot 5 + 7)^{\frac{3}{2}}} \\ a_2 = -\frac{1}{250} \\ \Rightarrow \frac{2}{2\sqrt{2 \cdot 5 + 7}} = \frac{18}{250} + a_1 \quad | + \frac{18}{250} \\ a_1 = \frac{34}{125} \\ \Rightarrow \sqrt{2 \cdot 5 + 7} = -\frac{81}{250} + \frac{9 \cdot 34}{125} + a_0 \quad | + \frac{81}{250} - \frac{9 \cdot 34}{125} \\ a_0 = \frac{715}{250} \\ \Rightarrow T_{2,5}(x) = \frac{-x^2 + 68x + 715}{250}$$

$$b) |T_{2,5}(x) - f(x)| \text{ für } x \in [8,5; 5,1] \\ \left| \int_{8,5}^{5,1} T_{2,5}(x) dx - \int_{8,5}^{5,1} f(x) dx \right| \\ \approx 1,6 \cdot 10^{-10} \\ 5,1 \cdot 8,5$$

$$c) \int_0^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx \quad | \text{Sub.} \\ u = \sqrt{x} \Rightarrow \sqrt{x} = u \quad | \cdot 2 \\ x = u^2 \quad | \frac{d}{du} \\ \frac{dx}{du} = 2u \quad | \cdot du \\ dx = 2u \cdot du \\ \Rightarrow \int_0^4 \frac{e^u}{u} \cdot 2u du = \int_0^4 2e^u du \quad | u = \sqrt{x} \\ \Rightarrow \int_0^4 2e^{\sqrt{x}} dx = [2e^{\sqrt{x}}]_0^4 = \dots$$

$$d) \int x^2 \cdot \sin(x) dx \quad | \text{Partielle Integration} \\ u = x^2 \Rightarrow u' = 2x \\ v' = \sin(x) \Rightarrow v = -\cos(x) \\ \Rightarrow -x^2 \cos(x) - \int 2x \cdot -\cos(x) dx \quad | \text{Partielle Int.} \\ u = 2x \Rightarrow u' = 2 \\ v' = -\cos(x) \Rightarrow v = -\sin(x) \\ \Rightarrow F(x) = -x^2 \cdot \cos(x) - (2x \cdot -\sin(x) - \int 2 \cdot -\sin(x) dx) \\ = -x^2 \cdot \cos(x) + 2x \cdot \sin(x) + 2 \cos(x) \quad \checkmark$$

$$e) \int |x^2 - 4| dx = \frac{(x^2 - 4)}{|x^2 - 4|} \cdot \int x^2 - 4 dx \\ \Rightarrow \int x^2 - 4 dx = \frac{1}{3} x^3 - 4x \\ \Rightarrow F(x) = \frac{(x^2 - 4)(\frac{1}{3} x^3 - 4x)}{|x^2 - 4|}$$

$$f) \int (6x^2 + 2x) \cdot \sqrt{-2x^3 - x^2 + 100} dx \quad | \text{Substitution} \\ u = -2x^3 - x^2 + 100 \quad | \frac{d}{dx} \\ \frac{du}{dx} = -6x^2 - 2x \quad | \cdot dx \cdot \frac{1}{-6x^2 - 2x} \\ dx = \frac{1}{-6x^2 - 2x} du$$

$$\int (\dots) \sqrt{u} dx = \int \frac{1}{5} \sqrt{u} du \\ = -\frac{2u^{\frac{3}{2}}}{3} \Rightarrow \frac{2(-2x^3 - x^2 + 100)^{\frac{3}{2}}}{-3} \rightarrow \dots$$

$$g) \int \frac{5x+8}{(1-x)^2} dx \quad | \text{Substitution} \\ u = 1-x \quad | \frac{d}{dx} \\ \Rightarrow x = 1-u$$

$$\frac{du}{dx} = -1 \quad | \cdot dx \quad | \cdot (-1) \\ dx = -1 \cdot du \\ \Rightarrow \int \frac{5(1-u)+8}{u^2} \cdot (-1) du \\ \Rightarrow \int \frac{5u+13}{u^2} du \\ = \int \frac{5}{u} du + \int \frac{13}{u^2} du \\ = \int 5u^{-1} du + \int 13u^{-2} du \\ = 5 \ln(|u|) - \frac{13}{u} \quad | \text{Resub.} \\ \Rightarrow 5 \ln(|1-x|) - \frac{13}{1-x}$$

$$h) \int \frac{\ln(\ln(x))}{x} dx \quad | \text{Subst.} \\ u = \ln(x) \quad | e^{\cdot} \quad | \frac{d}{dx} \\ e^u = x \\ \frac{du}{dx} = \frac{1}{x} \quad | \cdot dx \cdot x \\ dx = x \cdot du$$

$$\Rightarrow \int \frac{\ln(u)}{x} \cdot x du \\ \Rightarrow \int \ln(u) du \quad | \text{Standard Integral} \\ = u \cdot \ln(u) - u \quad | u = \ln(x) \\ \Rightarrow \ln(x) \cdot \ln(\ln(x)) - \ln(x)$$

$$5) f(x) = e^{x^2} \\ a) T_4(x) = f(x_0) + f'(x_0)(x-x_0) + \dots + \frac{f^{(4)}(x_0)}{4!} (x-x_0)^4$$

$$f(x) = e^{x^2} \\ f'(x) = 2xe^{x^2} \\ f''(x) = 2e^{x^2} + 4x^2 e^{x^2} = e^{x^2} (4x^2 + 2) \\ f'''(x) = 8xe^{x^2} + 2xe^{x^2} (4x^2 + 2) \\ = e^{x^2} (8x^3 + 12x)$$

$$f^{(4)}(x) = (24x^2 + 12) \cdot e^{x^2} + 2e^{x^2} (8x^3 + 12x) \\ = e^{x^2} (16x^4 + 48x^2 + 12)$$

$$\Rightarrow f(x_0) = f(1) = e \\ f'(1) = 2e \\ f''(1) = 6e \\ f'''(1) = 20e \\ f^{(4)}(1) = 76e$$

$$\Rightarrow T_{4,1}(x) = e \left(\frac{76}{4!} (x-1)^4 + \frac{20}{3!} (x-1)^3 + \frac{6}{2!} (x-1)^2 + 2(x-1) + 1 \right)$$

$$b) \left| \frac{\int_{0,8}^{1,2} T_{4,1}(x) dx - \int_{0,8}^{1,2} f(x) dx}{1,2 - 0,8} \right|$$

1a) $(\sqrt{2} + i) \cdot (8 - 3i) = (8 + \sqrt{2}) + 2i$

b) $(\sqrt{2} + i) \cdot (8 - 3i) = \sqrt{2} \cdot 8 - \sqrt{2} \cdot 3i + 8i + 3 = (8\sqrt{2} - 3\sqrt{2}) + (8 - 3)i = 5\sqrt{2} + 5i$

c) $z = \left(\frac{2}{1-i}\right)^2 = \frac{4}{(1-i)^2}$

$= \frac{4}{1-2i-1} = -\frac{2}{-i} = 2i$

$0 = z + \frac{2}{i}$

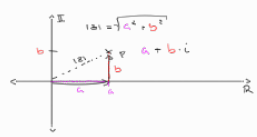
2a) zu beweisen: $|z| = \sqrt{2} \cdot |z^*|$

$z = a + bi$
 $z^* = a - bi$

$\Rightarrow I |a + bi| = \sqrt{(a+bi)(a-bi)}$
 $II |a + bi| = \sqrt{a^2 + b^2}$
 $I \Rightarrow \sqrt{a^2 - abi + abi + b^2} = \sqrt{a^2 + b^2}$
 $I = II$

b) zu beweisen / prüfen: $(z_1 \cdot z_2)^* = z_1^* \cdot z_2^*$

Nehm. Bed.: $z = a + bi$
 $\Rightarrow I (z_1 \cdot z_2)^* = ((a+bi)(c+di))^*$
 $II z_1^* \cdot z_2^* = (a-bi)(c-di)$
 $I (a^2 + 2abi - b^2)^* = (a^2 - b^2 + 2abi)^*$
 $\Rightarrow I a^2 - b^2 - 2abi$
 $II (a-bi)(c-di) = a^2 - 2abi - b^2 = a^2 - b^2 - 2abi$
 $I = II$
 $\Rightarrow (z_1 \cdot z_2)^* = z_1^* \cdot z_2^*$ gilt.



1) $E_0 \approx \frac{1}{2} z^2 = -3 + 4i$

mit: $z = a + bi$

$\frac{1}{2} z^2 = -3 + 4i \quad | \cdot 2$

$z^2 = -6 + 8i \quad | \sqrt{}$

$z = \pm \sqrt{-6 + 8i}$

$z \approx \pm \sqrt{10} \cdot e^{\frac{(n-\pi/2)}{2}}$

$z_1 \approx \sqrt{10} e^{\frac{2.44i}{2}} = 2.44i$

$z_2 \approx -\sqrt{10} e^{\frac{2.44i}{2}} = -2.44i$

u. z

$r = 2.44$
 $\varphi = 93.4^\circ \Rightarrow 0.83 \cdot 30.6$

3) a) $(8-i)(1-2i) = 8 - 16i - i + 2 = 6 - 17i$

b) $(1-i)^6 = ((1-i)^2)^3$

$= (1-2i-1)^3 = (-2i)^3 = 8i$

c) $z = \frac{2-i}{1-i} = (2-i) \cdot \frac{1}{1-i}$

$= \frac{2-i}{1-i} = \frac{2-i}{1-i} \cdot \frac{1+i}{1+i} = \frac{(2-i)(1+i)}{1-i^2} = \frac{2+2i-i-i^2}{2} = \frac{3+i}{2}$

$\Rightarrow |z| = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{9}{4} + \frac{1}{4}} = \sqrt{\frac{10}{4}} = \frac{\sqrt{10}}{2}$

2) $z^3 = -10 - 5i$

$-10 - 5i = 5\sqrt{5} e^{\frac{2.44i}{2}}$

$z_k = \sqrt[3]{5\sqrt{5}} e^{\frac{2.44i}{6}}$ (Moivre Formel)

$z_0 = 2.23 e^{\frac{2.44i}{6}}$

$z_1 = 2.23 e^{\frac{2.44i + 2\pi i}{6}}$

$z_2 = 2.23 e^{\frac{2.44i + 4\pi i}{6}}$

12.2) $g(x) = \cos(0.2x)$

$h(x) = x$

$f(x) = g(x) - h(x)$

$x_0 = 0.5$

$x_1 = 1$

1) Startwert: $a = \sqrt{3} = 3$

$b = \frac{11}{3}$

$\Rightarrow \frac{11}{3} + 3 = 3.3$

$\Rightarrow \frac{11}{3} + 3.3 = 3.16$

$\Rightarrow \frac{11}{3} + 3.3 = 3.16625 \rightarrow \dots \rightarrow 3.1662475$

b) $e^{i \cdot 3\pi} = e^{3\pi i} = -1$

$e^{i \cdot 3\pi} = -1$

oder: $r(\cos(\theta) + i \sin(\theta)) = re^{i\theta}$

12.3 Startwert: $x_0 = -1$

$f(x) = 4x^3 - 300x^2 - 0.16x + 12$

$f'(x) = 12x^2 - 600x - 0.16$

Newton-Verfahren: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$\Rightarrow x_1 = -1 + \frac{251.84}{611.84}$

$x_1 \approx -0.52$

$x_2 \approx -0.52 - \frac{-69.6}{345.4} \approx -0.3$

$x_3 \approx -0.3 - \frac{-14.34}{180.83} \approx -0.22$

$x_4 \approx -0.22 - \frac{-2.18}{134.16} \approx -0.2$ (Nullstelle von $f(x)$)

3) $r = -\sqrt{5} + 5i$

$s = -\sqrt{5} - 5i$

$z^4 = r + s = -2\sqrt{5} - 4i$

$z = \sqrt[4]{-2\sqrt{5} - 4i}$

$z_k = 1.57 e^{\frac{2.44i + 2\pi ki}{4}}$

$z = \left\{ 1.57 e^{\frac{2.44i + 2\pi ki}{4}} \mid k, n \in \mathbb{Z} \right\}$

Selbsterformel: $x_{n+1} = x_n - f(x_n) \cdot \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$

$\Rightarrow x_2 = 1 + 0.02 \cdot \frac{1 - 0.5}{-0.02 - 0.084}$

$x_2 = 0.5808$

12.3b) $20 \rightarrow -, 100 \rightarrow +$

$60 \rightarrow -, 100 \rightarrow +$

$80 \rightarrow +, 100 \rightarrow +$

$70 \rightarrow -, 80 \rightarrow +$

$75 \rightarrow 0 \rightarrow x_3 = 75$

12.4) $f(x) = 2e^{x^2}$

$T_1 \approx \int_0^1 2e^{x^2} dx$

$T_n = \frac{1}{2} \cdot \left(f(a) + 2 \cdot \sum_{i=1}^{n-1} f(a+i \cdot h) + f(b) \right)$

$h = \frac{b-a}{50} = \frac{1-0}{50} = \frac{1}{50}$

$\Rightarrow T_n = \frac{1}{100} \left(f(0) + 2 \cdot \sum_{i=1}^{49} f\left(i \cdot \frac{1}{50}\right) + f(1) \right)$

$T_n = \frac{1}{100} \left(2 + 2 \cdot \sum_{i=1}^{49} f\left(\frac{i}{50}\right) + 2e \right) \approx 2.52567 \approx 2.52530$

XI

a) $\mathcal{G}^T(144, 85)$

0. $a = 144, b = 85$

1. $a = 144 - 85 = 59$

2. $b = 85 - 59 = 26$

3. $a = 59 - 26 = 33$

4. $b = 33 - 26 = 7$

5. $a = 33 - 7 = 26$

6. $b = 26 - 7 = 19$

7. $a = 19 - 7 = 12$

8. $b = 12 - 7 = 5$

9. $a = 5 - 7 = -2$

10. $b = -2 - 7 = -9$

11. $a = -9 - 7 = -16$

$\Rightarrow \mathcal{G}^T(144, 85) = 1$

2a) a: Anzahl Vorkommen A, b: Anzahl Vorkommen B

$\Rightarrow 75a + 105b = 6000 \quad | -105b |$

$a = \frac{6000 - 105b}{75}$

Einsetzen von Werten:

$\dots \Rightarrow$ für $b=5 \rightarrow a = \frac{6000 - 105 \cdot 5}{75} = 73$

$\Rightarrow$ Ja, es ist möglich

b) Die Gleichung: $\left\{ b \in \mathbb{N} \mid \frac{6000 - 105b}{75} \% 1 = 0 \right\}$

bestimmt alle Möglichkeiten zur Lösung des Problems.

c) $a = 6732, b = 17282$

$a = a \% b = 11664$

$b = 17282 - 11664 = 5618$

$a = 11664 - 5618 = 6046$

$a = 6046 - 5618 = 428$

$b = 5618 \% 428 = 288$

$a = 528 \% 288 = 240$

$b = 288 \% 240 = 48$

$a = 240 \% 48 = 0$

$\Rightarrow \mathcal{G}^T(\dots) = 48$

12.4) $f(x) = 2e^{x^2}$

$T_1 \approx \int_0^1 2e^{x^2} dx$

$T_n = \frac{1}{2} \cdot \left(f(a) + 2 \cdot \sum_{i=1}^{n-1} f(a+i \cdot h) + f(b) \right)$

$h = \frac{b-a}{50} = \frac{1-0}{50} = \frac{1}{50}$

$\Rightarrow T_n = \frac{1}{100} \left(f(0) + 2 \cdot \sum_{i=1}^{49} f\left(i \cdot \frac{1}{50}\right) + f(1) \right)$

$T_n = \frac{1}{100} \left(2 + 2 \cdot \sum_{i=1}^{49} f\left(\frac{i}{50}\right) + 2e \right) \approx 2.52567 \approx 2.52530$

Prüfung

1) a) $a_1 = -\frac{3}{2}, a_2 = -\frac{7}{4}, a_3 = -\frac{15}{8}$

$a_4 = -\frac{31}{16}, a_5 = -\frac{63}{32}$

b) Funktion $f(x)$ gilt $f(-2) = 0$, d.h. sie ist gegen -2 konvergent.

c) Die Funktion ist von -1 bis -2 beschränkt.

$-1 \geq f(x) > -2$

d) Zeigen, dass die Steigung der Funktion immer negativ bleibt, ist die Funktion monoton fallend.

2) $\sum_{n=1}^{\infty} \frac{n}{3^n}$

In TR mit Oberer Grenze 5555 eingesetzt:

$\Rightarrow \frac{1}{3} + \frac{2}{3} + \frac{3}{27} + \dots \Rightarrow \frac{3}{4}$

$\Rightarrow$ Die Reihe konvergiert gegen $\frac{3}{4}$

3) $z_1^* = 2 + i$

$\Rightarrow \frac{1}{z_1} + z_2^* \cdot (z_1 - z_2) = \frac{1}{1+i} + (2+i) \cdot (1-i-2-i)$

$= \frac{1}{1+i} + 2 + 2i - 4 + 2i + i - 1 - 2i - 1$

$= \frac{1}{1+i} - 4 + 3i$

$= \frac{1-i}{(1+i)(1-i)} - 4 + 3i$

1) $a_1 = -2$ $a_2 = -2,5$ $a_3 = -2,75$

a) $a_4 = -2,875$ $a_5 = -2,9375$

b) $a_n > a_{n+1}$, da: $a_{n+1} = a_n - \frac{1}{2^n}$

ergibt sich: Die Folge ist auf $[-3; -1]$ beschränkt.

c) $a_n > a_{n+1}$, da: $a_{n+1} = a_n - \frac{1}{2^n}$
und $\frac{1}{2^n} > 0$, da $\frac{1}{x} \neq 0$ & $2^n > 0, n \in \mathbb{R}$

=> Die Folge ist für die Reellen Zahlen streng monoton fallend, für die Komplexen Zahlen aber nicht, da: $-\infty < 2^n < \infty, n \in \mathbb{C}$

d) aus $-1 - \sum_{n=0}^{\infty} \frac{1}{2^n} = -3$ folgt, dass die Folge gegen -3 konvergiert.

2) $\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{3}{4}$

Die Reihe konvergiert gegen $\frac{3}{4}$.

3) $z_1 = 1 \cdot i$, $z_2 = 2 \cdot i$, $z_2^* = 2 \cdot i$

=> $\frac{1}{z_1} + z_2^* \cdot (z_1 - z_2)$

= $\frac{1}{1 \cdot i} + (2 \cdot i) \cdot (1 \cdot i - 2 \cdot i)$

= $\frac{1-i}{(1+i)(1-i)} + (2 \cdot i) \cdot (-1 \cdot 2 \cdot i)$

= $\frac{1-i}{2} + (-2 \cdot 4 \cdot i \cdot -2)$

= $\frac{1}{2} - \frac{i}{2} - 4 + 3i = -3,5 + 2,5i$

Realteil: $-3,5$ Imaginärteil: $2,5i$

b) $|z_1| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$ Betrag

Argument: $\arctan\left(\frac{-1}{-1}\right) = \frac{5}{4}\pi$

c) $(-z_1)^{50} = ((-(-1-i))^5)^8$

$(-z_1)^2 = 1 + 2 \cdot i \cdot -1 = 2i$

$(2i)^5 = 32i$

$(32i)^5 = 33554432i$

4) $f(x) = x \cdot \ln(x)$

a) $\omega = [0; \infty]$, $D = [0; \infty]$

Nulstellen: $x \cdot \ln(x) = 0$ | Satz des Nullprodukts

$\underline{x_1 = 0}$ $\wedge$ $\ln(x) = 0$ | $e^0 = 1$
 $\underline{x_2 = e^0 = 1}$

Symmetrie:

V-Achsen-Symmetrie: $f(-x) = f(x) = -x \cdot \ln(-x) + x \cdot \ln(x)$
=> nicht gegeben.

Parabelsymmetrie zum Ursprung: $f(-x) = -f(x) = -x \cdot \ln(-x) \neq -x \cdot \ln(x)$
=> nicht gegeben.

$\lim_{x \rightarrow \infty} x \cdot \ln(x) \rightarrow \infty \cdot \ln(\infty) = \infty$

$\lim_{x \rightarrow 0} x \cdot \ln(x) \rightarrow 0 \cdot \ln(0^+) = 0$

b) $f'(x) = 0$

$f'(x) = u \cdot v' + u' \cdot v$

$u = x$, $v = \ln(x)$

$u' = 1$, $v' = \frac{1}{x}$

$f'(x) = x \cdot \frac{1}{x} + 1 \cdot \ln(x) = 1 + \ln(x)$

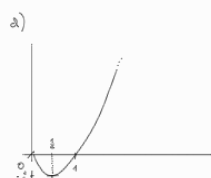
=> $0 = 1 + \ln(x)$ | -1 | e^0

$\frac{1}{e} = x$

Hinweis: $f'(x) = 0$

$f'(x) = \frac{1}{x}$, $f'(\frac{1}{e}) = e$

=> $0 \Rightarrow \tau \Rightarrow \text{bei } (\frac{1}{e}, f(\frac{1}{e})) = (\frac{1}{e}, -\frac{1}{e})$



4) $f(x) = \ln(x)$

a) $\text{Nohr. Bedingung: } f(x) = 0$

$0 = x \cdot \ln(x)$ | Satz des NP

$\underline{x_1 = 0}$ $\vee$ $\ln(x) = 0$ | $e^0 = 1$
 $\underline{x_2 = e^0 = 1}$

V-Achsen-Symmetrie: $f(x) = f(-x)$?

$x \cdot \ln(x) \neq -x \cdot \ln(-x)$

=> keine V-Achsen-Symmetrie

Parabelsymmetrie: $f(-x) = -f(x)$?

$-x \ln(-x) \neq -x \ln(x)$

=> keine Parabelsymmetrie

x-Achsen-Symmetrie: $f(x) = -f(x)$?

$x \ln(x) \neq -x \ln(-x)$

=> keine x-Achsen-Symmetrie

$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \ln(x) \rightarrow \infty$

$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} 0 \cdot \ln(0^+) \rightarrow 0$

b) $f'(x) = 0$

$f'(x) = u \cdot v' + u' \cdot v$

$u = x$, $v = \ln(x)$

$u' = 1$, $v' = \frac{1}{x}$

=> $f'(x) = x \cdot \frac{1}{x} + 1 \cdot \ln(x) = \ln(x) + 1$

Nohr. Bed: $f'(x) = 0$

=> $0 = \ln(x) + 1$ | -1

$-1 = \ln(x)$ | e^0

$\frac{1}{e} = x$ ✓

$f'(\frac{1}{e} + 0,01) = 0,03$, also positiv

=> TP bei $(\frac{1}{e}, f(\frac{1}{e})) = (\frac{1}{e}, -\frac{1}{e})$

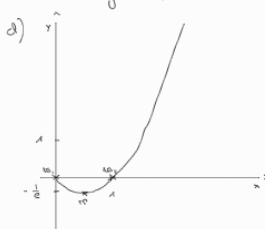
dadurch dass $\ln(x) + 1$ nur 1 Lösung im Reellen Raum hat, ist das die einzige Lösung -> kein HP

c)

$f''(x) = \frac{1}{x}$, Nohr. Bedingung: $f''(x) = 0$

$0 = \frac{1}{x}$ ✗

keine Lösungen für x => keine Wendepunkte



7)

$f(x) = 2x^2$, $g(x) = x^3 \cdot 3x$

$P_1 = (x_1, y_1)$

$P_2 = (x_2, y_2)$

$P_3 = (x_3, y_3)$

a) Ich nehme an: $x_1 \leq x_2 \leq x_3$

$\int_{x_1}^{x_3} |f(x) - g(x)| dx$

b)

Schrittprodukt: $f(x) = g(x)$

$x^3 - 3x = 2x^2$ | $-2x$

$x^3 - 2x^2 - 3x = 0$ | Satz des NP

$x_1 = 0$ $\vee$ $x^2 - 2x - 3 = 0$ | pq-Formel

$x_2 = 1 + \sqrt{1+3} = 3$ $= 281,64$

$x_3 = 1 - \sqrt{1+3} = -1$

=> $\int_{-1}^3 |x^3 - 2x^2 - 3x| dx = \left[\frac{1}{4}x^4 - \frac{2}{3}x^3 - \frac{3}{2}x^2 \right]_{-1}^3$

= $\frac{34}{6}$ FE

8) $f(x) = x^2 + \pi + \sin(x + \pi)$, $f(c) \approx 13,1$

$f'(x) = 2x + \cos(x + \pi)$, $f'(c) \approx 7,28$

$f''(x) = 2 - \sin(x + \pi)$, $f''(c) = 2$

$f'''(x) = -\cos(x + \pi)$, $f'''(c) = -1$

$T_{n,c} = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} \cdot (x-c)^k$

$T_{3,c}(x) = (x-c)^0 \cdot \frac{f(c)}{0!} + (x-c)^1 \cdot \frac{f'(c)}{1!} + (x-c)^2 \cdot \frac{f''(c)}{2!} + (x-c)^3 \cdot \frac{f'''(c)}{3!}$

= $\frac{-(x-c)^3}{6} + (x-c)^2 + 7,28(x-c) + 13,1$

$= \frac{1-i}{1-i} \cdot \frac{1-i}{1-i} + 1 - 4 + 3i$

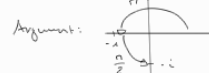
= $\frac{1}{2} - \frac{i}{2} - 4 + 3i = 3,5 + 2,5i$

$\mathbb{R}: 3,5$, $\mathbb{I}: 2,5i$

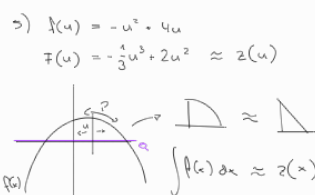
b) $|-z_1| = |-1-i|$

=> $\sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$

$\theta(-z_1) = \frac{3\pi}{2}$



c) $(-1-i)^{50} = 33554432i$



b)

HP von $z(u)$:

$z'(u) = -\frac{3}{2}u^2 + 4u$

Nohr. Bed.: $z'(u) = 0$

$0 = -\frac{3}{2}u^2 + 4u$ | $\cdot (-\frac{2}{3})$

$0 = u^2 - \frac{8}{3}u$ | pq-Formel

$x_1 = \frac{8}{6} + \sqrt{\frac{64}{9}} = 4$

$x_2 = \frac{8}{6} - \sqrt{\frac{64}{9}} = -\frac{4}{3}$

$z''(u) = -3u - 4$

$z''(4) = -8$, $z'(-\frac{4}{3}) = 8$

=> HP bei $x = -\frac{4}{3}$

=> Maximales Flächeninhalts von Δ

$\hookrightarrow -\frac{1}{2} \cdot (-\frac{4}{3})^2 + 2 \cdot (-\frac{4}{3})^2$
 $= 4,74 \text{ FE}$

6)

a) $\int \cos(2c + bx) \cdot \frac{4}{2c + bx}$

=> $\frac{\sin(2c + bx)}{b} + \frac{4 \ln|2c + bx|}{b}$

b)

$\int (4x^3 - 1) \cdot \ln(x) dx$ | Partielle

$u = \ln(x)$ => $u' = \frac{1}{x}$

$v = 4x^3 - 1$ => $v' = 12x^2$

=> $F(x) = u \cdot v - \int u' \cdot v dx$

=> $\ln(x) \cdot (4x^3 - 1) - \int \frac{1}{x} \cdot (4x^3 - 1) dx$

=> $x^4 \ln(x) - x \ln(x) - \int x^3 - \frac{1}{x} dx$

=> $x^4 \ln(x) - x \ln(x) - \frac{1}{4}x^4 + x$

$\int_2^4 x^4 \ln(x) - x \ln(x) - \frac{1}{4}x^4 + x$

= $\frac{4}{5}x^5 \ln(x) - \frac{1}{2}x^2 \ln(x) - \frac{1}{20}x^5 + \frac{1}{2}x^2$ ✓

b) $\int_0^{\frac{\pi}{2}} |T_{3,n}(x) - f(x)| dx = (\dots)$

8) a) $\int (\cos(2x + b \cdot x) + \frac{4}{2x-10}) dx$
 $= \int \cos(2x + bx) dx + 4 \cdot \int \frac{1}{2x-10} dx$ | Substitution
 $u = 2x + bx, v = 2x - 10$
 $\int \cos(u) du + 4 \cdot \int \frac{1}{v} dv$
 $du = \frac{dx}{1} = \frac{1}{1} dx, dv = \frac{dx}{1} = \frac{1}{1} dx$
 $= \frac{\sin(2x + bx)}{1} + \frac{4 \ln|2x - 10|}{1}$

b) $\int_2^4 ((4x^3 - 1) \cdot (\ln(x))) dx$ | Partielle Integration
 $u = \ln(x) \Rightarrow u' = \frac{1}{x}$
 $v' = 4x^3 - 1 \Rightarrow v = x^4 - x$

$\Rightarrow u \cdot v - \int u' \cdot v dx$
 $= x^4 \ln(x) - x \ln(x) - \int \frac{1}{x} (x^4 - x) dx$
 $= \dots - \int x^3 - 1 dx$
 $= x^4 \ln(x) - x \ln(x) - \frac{1}{4} x^4 + x$

5) $f(x) = (x+1) \cdot \sin(x) + \cos(x)$
 $f'(x) = \frac{d}{dx} [(x+1) \cdot \sin(x)] + \frac{d}{dx} [\cos(x)]$
 $u = x+1 \Rightarrow u' = 1$
 $v = \sin(x) \Rightarrow v' = \cos(x)$
 $f'(x) = (x+1) \cdot \cos(x) + \sin(x) - \sin(x)$
 $f'(x) = (x+1) \cos(x)$
 Notw. Bed.: $f'(x) = 0$

$0 = (x+1) \cos(x)$ | Satz des NP
 $x+1 = 0 \vee \cos(x) = 0$ | $\arccos()$
 $x_1 = -1, x_2 = \frac{\pi}{2} + 2\pi k, k \in \mathbb{Z}$
 d.h.: Extremstellen in $[0; 10]$
 $= \left\{ \frac{\pi}{2}; \frac{5\pi}{2} \right\}$

6) a) $I = [0; 100] \in \mathbb{R}$
 Gewinn: $G(x) = -x^4 + 180x^3 - 3500x^2, x \in I$
 Haltepunkte: $G'(x) = 0, x \in I$
 $G'(x) = -4x^3 + 540x^2 - 7000x$
 Notw. Bedingung: $G'(x) = 0$
 $0 = -4x^3 + 540x^2 - 7000x$ | Satz des NP
 $x_1 = 0, -4x^2 + 540x - 7000 = 0 \vee -4$
 $x_2 = \frac{540}{8} = 67.5, x_3 = \frac{7000}{8} = 875$
 $x_3 = \frac{7000}{8} - \sqrt{19225 - 2750} = 110$
 $x_3 = \frac{7000}{8} - \sqrt{19225 - 2750} = 25$

dadurch, dass $x=0$ kein Sinn macht und Ansatz kein MoG hat, sollte er 25g herstellen.

b) In diesem Fall sollte er natürlich MoG herstellen.

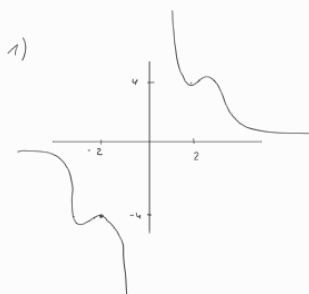
7) Fläche: $\int_{-1}^1 |f(x) - g(x)| dx$
 $\Rightarrow \int_{-1}^1 |32x^2 - e^{2x} + \sin(x) - 4 + e^{2x} - \cos(x - \frac{\pi}{2})| dx$
 $\sin(x) = \cos(x - \frac{\pi}{2})$
 $\Rightarrow \int_{-1}^1 32x^2 dx = \left[\frac{32}{3} x^3 \right]_{-1}^1 = \left| \frac{32}{3} \cdot 1 - \frac{32}{3} \cdot (-1) \right| = \frac{64}{3} FE$

8) $\lim_{n \rightarrow \infty} \sum_{i=2}^{\infty} 3 \cdot \left(\frac{1}{5}\right)^i$
 TR:
 $\sum_{i=2}^{\infty} 3 \cdot \left(\frac{1}{5}\right)^i = \frac{3}{20}$
 $\Rightarrow$ Der Grenzwert der Reihe ist $\frac{3}{20}$

7) a) $\int_1^2 (f(x) - g(x)) dx$
 b) Schnittpunkte: $f(x) = g(x)$
 $\Rightarrow 2x^2 = x^3 - 2x^2 - 3x$ | $-2x^2$
 $0 = x^3 - 2x^2 - 3x$ | Satz des NP
 $x = 0, 1 = x^2 - 2x - 3$ | PQ-Formel
 $x_1 = 1 + \sqrt{1+3} = 3$
 $x_2 = 1 - \sqrt{1+3} = -1$
 $\Rightarrow A_{\text{Fläche}} = \int_{-1}^3 x^3 - 2x^2 - 3x dx = \frac{74}{6} FE$

8) a) $T_{3,n}(x)$:
 $f(x) = x^2 + \sin(x, n) \cdot \pi$
 $f'(x) = 2x + \cos(x, n) \cdot \pi$
 $f'(x) = 2 - \sin(x, \pi)$
 $f''(x) = -\cos(x, \pi)$
 $T_{3,n}(x) = \sum_{k=0}^3 \frac{f^{(k)}(n)}{k!} \cdot (x-n)^k$
 $T_{3,n}(x) \approx \frac{-1}{6} (x-n)^3 + (x-n)^2 + 7.28(x-n) + 13.04$

b) Lagrange-Polynom:
 $f^{(n)}(x) = \sin(x, \pi)$
 $\max_{x \in [0, 2\pi]} |\sin(x, \pi)| = 1$
 $\Rightarrow R_3(x) = \frac{1}{4!} (x-n)^4$
 $= \frac{(\frac{15}{24})^4}{24} = 0.02$



2) $\int_0^{\frac{\pi}{2}} \frac{\cos(x)}{1 + 2\sin(x)} dx$ | Substitution
 $u = 2\sin(x) + 1$
 $du = \frac{dx}{u} = \frac{dx}{2\cos(x)}$
 $\int \frac{1}{u} du = \ln|u|$ | Rücksubstitution
 $\Rightarrow \frac{\ln(1 + 2\sin(x) + 1)}{2}$
 $= \frac{\ln(2 + 2\sin(x))}{2}$

3) SP: = Schnittpunkt von f & g
 a) $\int_1^2 |f(x) - g(x)| dx$
 b) SP berechnen: $f(x) = g(x)$
 $\Rightarrow \sqrt{x(3-x)} = 2x$
 $\sqrt{x(3-x)} = 2x$ | Satz des NP
 $\sqrt{x} = 2\sqrt{3-x}$
 $\sqrt{x} = 0 \vee 3-x = 2\sqrt{x}$ | Potenzen mit $x=1$
 $x_1 = 0, 3-x = 2\sqrt{x}$
 $3-x = 2\sqrt{x}$
 $\Rightarrow x_2 = 2$
 $\Rightarrow \int_0^2 \sqrt{x(3-x)} - 2x dx \approx 1.81 FE$

Satz

1) a) gebrochen Rational
 b) Exponential
 c) $\frac{1}{x}$
 d) Wurzel
 e) Grenzwert
 2) a) $f_u(x) = 0$
 $f_u(x) = e^{4x} \frac{2x^2 - 4x - 26}{(x^2 + 3x - 20)} = 0$ | Satz des NP
 $2x^2 - 4x - 26 = 0 \vee \frac{1}{(x^2 + 3x - 20)} \neq 0, x \in \mathbb{R}$
 $x^2 - 2x - 13 = 0$ | PQ
 $x_1 = 1 + \sqrt{1+13}$
 $x_2 = 1 - \sqrt{1+13}$

Polstellenberechnung:
 Nenner = 0:
 $(x^2 + 3)(x - 20) = 0$ | Satz des NP
 $x^2 + 3 = 0 \vee x - 20 = 0$ | $x=20$
 $x^2 = -3 \vee \sqrt{-3} \quad \underline{x_3 = 20}$
 $\Rightarrow x_1, x_2 \notin \mathbb{R}$
 $\Rightarrow$ Die einzige Polstelle ist bei $x = 20$

3) $\frac{\cos(0.6x + 2)}{\cos(x)} = \frac{1}{2} \tan(x)$ | $\cos(x) = \frac{1}{\tan(x)}$
 $\Rightarrow \cos(0.6x + 2) \cdot \tan(x) = \frac{1}{2}$ | Satz des NP
 $x_1 = 0 + 2\pi k, k \in \mathbb{Z}$
 $x_2, \dots: \cos(0.6x + 2) = \frac{1}{2}$ | $\arccos()$
 $0.6x + 2 = \arccos(\frac{1}{2}) + 2\pi k \vee 2 \cdot \frac{\pi}{3}$
 $x_2 = \frac{(\arccos(\frac{1}{2}) - 2 + 2\pi k) \cdot 5}{3}, k \in \mathbb{Z}$
 $x_3 = \frac{(2\pi - \arccos(\frac{1}{2}) - 2 + 2\pi k) \cdot 5}{3}, k \in \mathbb{Z}$

4) a) $N(t) = N_0 e^{-0.07t}$
 $N'(t) = -0.07 \cdot N_0 e^{-0.07t}$
 Notw. Bedingung: $N'(t) = 0$
 $0 = -0.07 \cdot N_0 e^{-0.07t}$ | Satz des NP
 $-0.07 \cdot N_0 \neq 0 \vee e^{-0.07t} = 0$ | $\ln()$
 Aus dem Kontext heraus muss es kein Sin, dass die Anfangsmenge 0 ist $\Rightarrow e^{-0.07t} > 0$
 $\Rightarrow$ keine Extremstelle

$N''(x) = 0.07^2 N_0 \cdot e^{-0.07t} \neq 0$
 $\Rightarrow$ keine Wendestelle, Beweis wie oben / teilen
 b) $\frac{27}{3} = 27 \cdot e^{-0.07t}$ | $\div 27$
 $\frac{1}{3} = e^{-0.07t}$ | $\ln()$
 $\ln(\frac{1}{3}) = -0.07t$ | $\vee (-0.07t)$
 $\underline{15.69 \approx t}$
 A: Nach 15.69 Tagen ist weniger als ein Drittel der Substanz vorhanden.
 c) 1. Wenn es 7 Tage
 $2 = N_0 \cdot e^{-0.07 \cdot 7}$ | $\div e^{-0.07 \cdot 7}$
 $3.26 \approx N_0$

A: Der Forscher misst 3.26g Alster lach
 d.h. Anfangsmenge misst

4) $f(x) = \ln(x+2) + x^2 + 1$
 a) $T_{3,3}(x) = \frac{f''(3)}{3!} (x-3)^3 + \frac{f'(3)}{2!} (x-3)^2 + f'(3) \cdot (x-3) + f(3)$
 $f'(x) = \frac{1}{x+2} + 2x$
 $f''(x) = -\frac{1}{(x+2)^2} + 2$
 $f'''(x) = \frac{2x+4}{(x+2)^3}$
 $T_{3,3}(x) = \frac{3}{750} (x-3)^3 + \frac{44}{250} (x-3)^2 + 6.2(x-3) + 11.64$
 b) $\int_3^4 |T_{3,3}(x) - f(x)| dx$
 $\frac{\int_3^4 |T_{3,3}(x) - f(x)| dx}{4-3} \approx 0.32731$
 $\Rightarrow$ Der durchschnittliche Unterschied zwischen f & g liegt bei ≈ 0.32731

5)

$$z^3 := u - 4v$$

$$u := 4 + 57i$$

$$v := 1 + 100i$$

$$\Rightarrow z^3 = 4 + 57i - 4(1 + 100i)$$

$$z^3 = 4 + 57i - 4 - 400i$$

$$z^3 = -343i \rightarrow \text{wertiges Produkt}$$

Exponentialdarstellung: $343e^{\frac{3\pi}{2}i}$

$$z^3 = 343e^{\frac{3\pi}{2}i + 2\pi ki}, k \in \mathbb{Z} \mid 1 \frac{1}{3}$$

$$z = 343e^{\frac{3\pi}{6}i + \frac{2}{3}\pi ki}, k \in \mathbb{Z}$$